

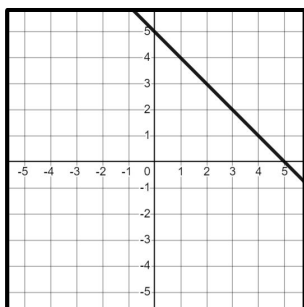


The equation of a straight line

Task A: Different forms of linear relationships

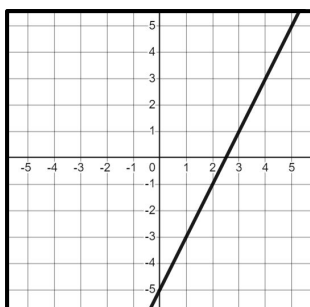
1) Plot each equation and identify the gradient and y-intercept.

a) $x + y = 5$



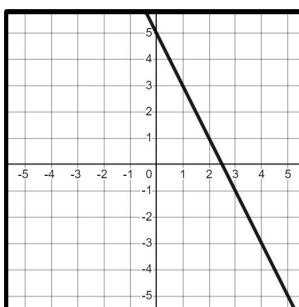
gradient = -1
y-intercept = $(0, 5)$

b) $y = 2x - 5$



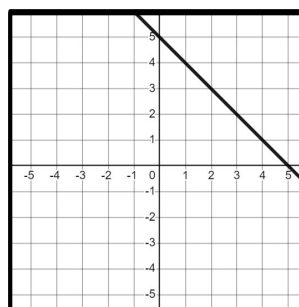
gradient = 2
y-intercept = $(0, -5)$

c) $y = 5 - 2x$



gradient = -2
y-intercept = $(0, 5)$

d) $2x + 2y = 10$



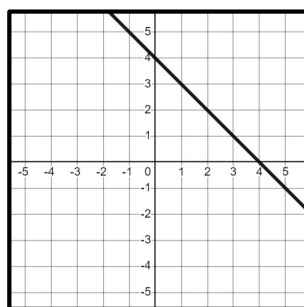
gradient = -1
y-intercept = $(0, 5)$

2) Match up any equations that represent the same relationship

a) and d)

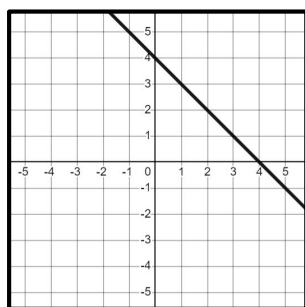
3) Plot each equation and identify the gradient and y-intercept.

a) $x + y = 4$



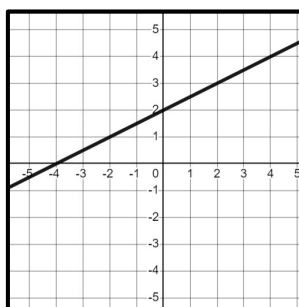
gradient = -1
y-intercept = $(0, 4)$

b) $y = 4 - x$



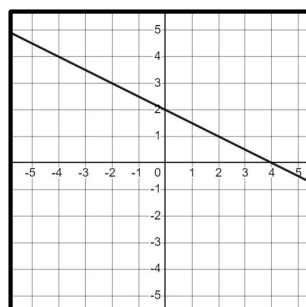
gradient = -1
y-intercept = $(0, 4)$

c) $y = \frac{1}{2}x + 2$



gradient = $\frac{1}{2}$
y-intercept = $(0, 2)$

d) $2y + x = 4$



gradient = $-\frac{1}{2}$
y-intercept = $(0, 2)$

4) Match up any equations that represent the same relationship

a) and b)



Task B: Gradient and y-intercept from $y = mx + c$

1) Fill in the table with the gradient and y-intercept for each equation.

Write the y-intercept as a coordinate.

equation	gradient	y-intercept
a) $y = 8x + 4$	8	$(0, 4)$
b) $y = 3x - 7$	3	$(0, -7)$
c) $y = -5x + 1$	-5	$(0, 1)$
d) $y = 10 - 3x$	-3	$(0, 10)$
e) $y = x$	1	$(0, 0)$
f) $y = -x$	-1	$(0, 0)$
g) $y = 5$	0	$(0, 5)$
h) $y = p + (q + 3)x$	$q + 3$	$(0, p)$

2) Fill in the missing pieces of information for each question.

equation	gradient	y-intercept
a) $y = 9x + 4$	9	$(0, 4)$
b) $y = \frac{1}{2} - 6x$	-6	$(0, \frac{1}{2})$
c) $y = 3(4x + 2)$	12	$(0, 6)$
d) $y = 10 - x$	-1	$(0, 10)$
e) $5(1 - 2x) = y$	-10	$(0, 5)$
f) $8 - 5x = y$	-5	$(0, 8)$
g) $y = u + tx$	t	$(0, u)$

